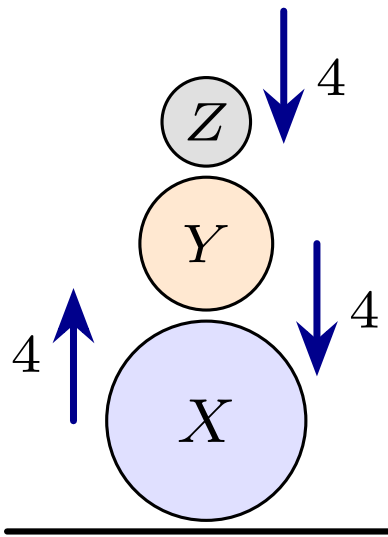
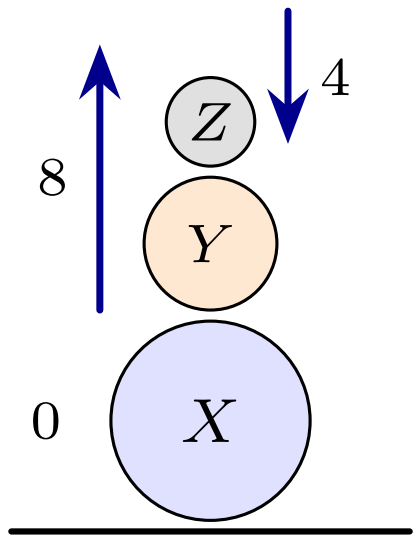
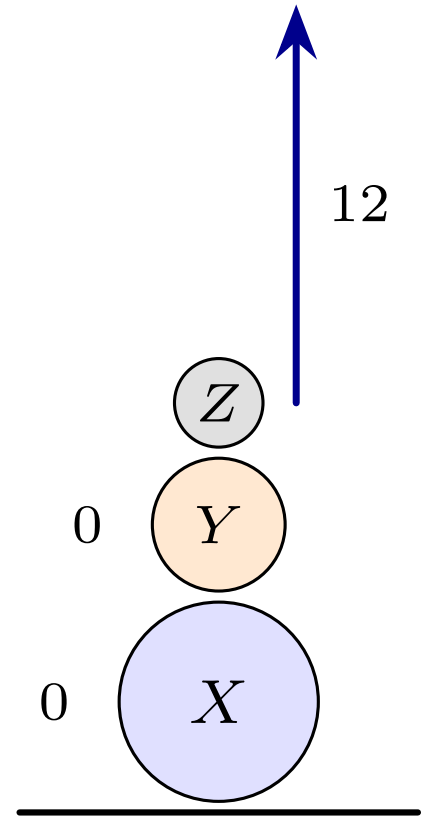
speeds in m s^{-1}; arrows show direction



after X bounces



after X hits Y



after Y hits Z